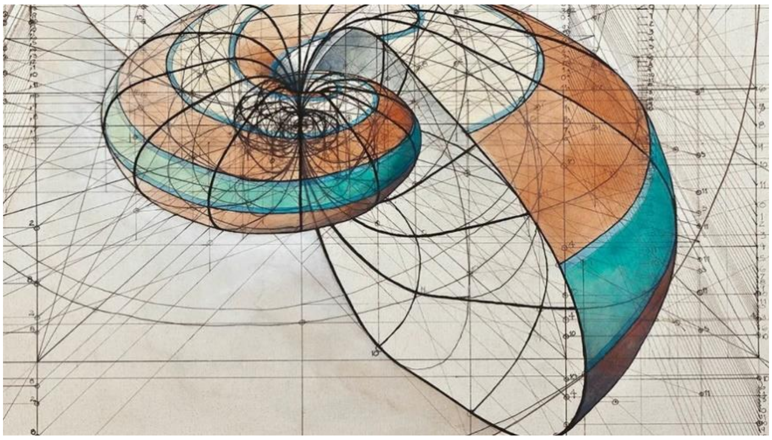
Tour de Maths 2023

Leg 2

Question Paper



The Golden Ratio

With special thanks to the sponsorship of

CASIO

SECTION A: 5 MARKS EACH

Question 1

The value of $2023^2 – 2022^2$ is:

A: 4092528   B: 1   C: 5   D: 4045

Question 2

In a certain code, if HAND is written as SZMW, then MILK is written as:

A: NROP   B: NOPR   C: NORP   D: RNOP

Question 3

If m is an odd integer and r is an even integer, which of the following is an even integer?

A: $m + r$ B: $mr + m$ C: $mr - r$ D: $m^2 - r$

Question 4

A cricket team (11 players) scores 383 runs.

If the average score of the first 6 players was 48 runs.

What was the average score of the last 5 batsmen?

A: 19	B: 22	C: 25	D: 27

Question 5

Giles numbers are multiples of all the numbers 1, 2, 3, 4, 5, 6, 7, 8, 9, and 10.

An example is $10! = 3\ 628\ 800$

What is the smallest *Giles* number?

A: 2520	B: 1260	C: 3720	D: 3972

Question 6

Forty-five has an interesting property: the sum of its digits is the same as number of letters (excluding the hyphen)

How many numbers from 1 to 45, including 45, have this property?

A: 7	B: 8	C: 9	D: 10

Question 7

You are given five weights. 1kg, 2kg, 5kg, 6kg, 10kg.

How many different weights can you obtain using these weights?

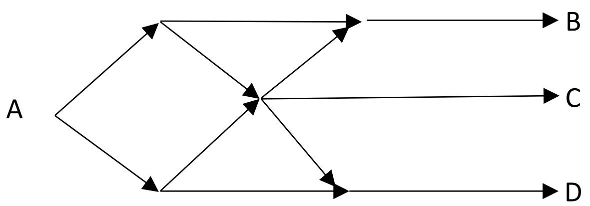
A: 16 B: 25 C: 26 D: 31

Question 8

An ant starts at A on the grid of each of the one-way paths shown.

When it reaches a junction is has an equal probability chance to take the next segment.

What is the probability that it will reach C?



A: 1/2 B: 1/3 C: 1/4 D: 1/6

Question 9

The printer at our school was attacked by a friendly virus.

All the holey letters have their holes filled in!

Each ink refill is expected to print exactly one million letters.

With the virus, the holey letters now use three times the ink: one for the normal part of the letter and two for the hole.

A new refill is put into the printer to print the following phrase until the refill is empty.

Quarter finals of the tenth mathematical and logical championship

What will the last letter printed be?

A: f   B: t   C: h   D: e

Question 10

Tom is looking at an encyclopaedia.

The first four pages are unnumbered pages followed by 256 interior pages numbered from 1 to 256.

The left pages have even numbers, and the right pages have odd numbers.

Tom opens the encyclopaedia at a random page.

He **adds up the six digits** of the two-page numbers in front of him.

This sum is the greatest it can be.

What is the number on the left-hand page?

A: 190 B: 194 C:198 D: 202

SECTION B: 10 MARKS EACH

Question 11

A three-digit number $4x2$ is added to the number 127 to give the three-digit number $5y9$.

If $5y9$ is exactly divisible by 9, then $x + y$ equals:

Question 12

A father's age is equal to the sum of the ages of his 5 children.

After 15 years, his age will only be half the sum of the ages of his 5 children.

How old is the father?

Question 13

Milan walks 10m North. From this position, he then walks 6m South. Thereafter, he walks 3m East. How far, and what direction is he in reference to his starting point?

Question 14

A car takes three hours to reach a destination by traveling at 60km/h. How long would the same trip take if they were travelling at 90km/h?

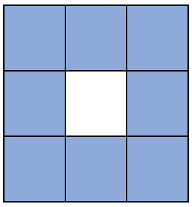
Question 15

Kevin is afraid of making an error in math class so in place of a calculator he carries a multiplication table with him.

X	2	3	4	5	6	7	8	9
2	4	6	8	10	12	14	16	18
3	6	9	12	15	18	21	24	27
4	8	12	16	20	24	28	32	36
5	10	15	20	25	30	35	40	45
6	12	18	24	30	36	42	48	54
7	14	21	28	35	42	49	56	63
8	16	24	32	40	48	56	64	72
9	18	27	36	45	54	63	72	81

His older brother makes a square mask with a square hole in the middle.

The mask fits exactly over 9 numbers.



Matthew told Kevin that the sum of the eight numbers covered by the mask was 288.

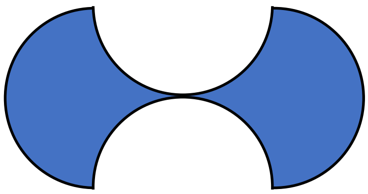
What was the centre number?

SECTION C: 20 MARKS EACH

Question 16

The figure shown is bounded by four semicircles each of radius r.

Calculate the shaded area.



Question 17

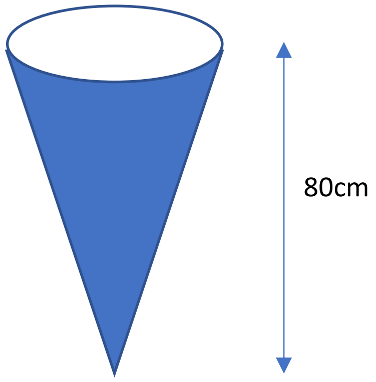
The formula for the volume of a cone is: $V = \frac{1}{3}\pi r^2 h$

A container in the shape of a right circular cone measuring 80cm from base to apex.

It holds a precious liquid.

When it is filled to the height of 20cm the value is R5.

What is the liquid worth when the cone is filled completely?



Question 18

Beena and Siya share a bag of sweets.

Beena takes one sweet the Siya takes two, then Beena takes three and Siya takes four, and so on, until the sweets are finished.

Siya is the last to take and the bag is empty.

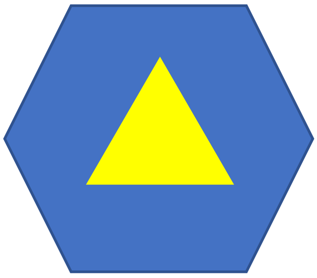
Siya then has 10 more sweets that Beena.

How many sweets were in the bag?

Question 19

A regular hexagonal nut has an equilateral triangle cut out of it.

If the sides of the hexagon and the triangle are equal in length, what is the ratio of the blue area to the area of the triangle?



Not drawn to scale

Question 20

Everybody in a room shakes hands with everybody else.

The total number of handshakes is 300.

What was the total number of people in the room?